

# S01 System boundary and sheet map

Central Board Rev.1 | 2026-09-26 V2 harness and integrated coil driver | manual KiCad transcription reference



- S02A 5V input, eFuse and USB power sharing
- S02B six 5V branches and Teensy socket pin map
- S02M Teensy mechanical placement and pin orientation
- S03A/B CAN common circuit and connector mapping
- S04A/B E-stop diagnostic circuit and integrated E228 coil driver
- S05 external I/O general circuit and connector map
- S11 unresolved decisions and transcription checks

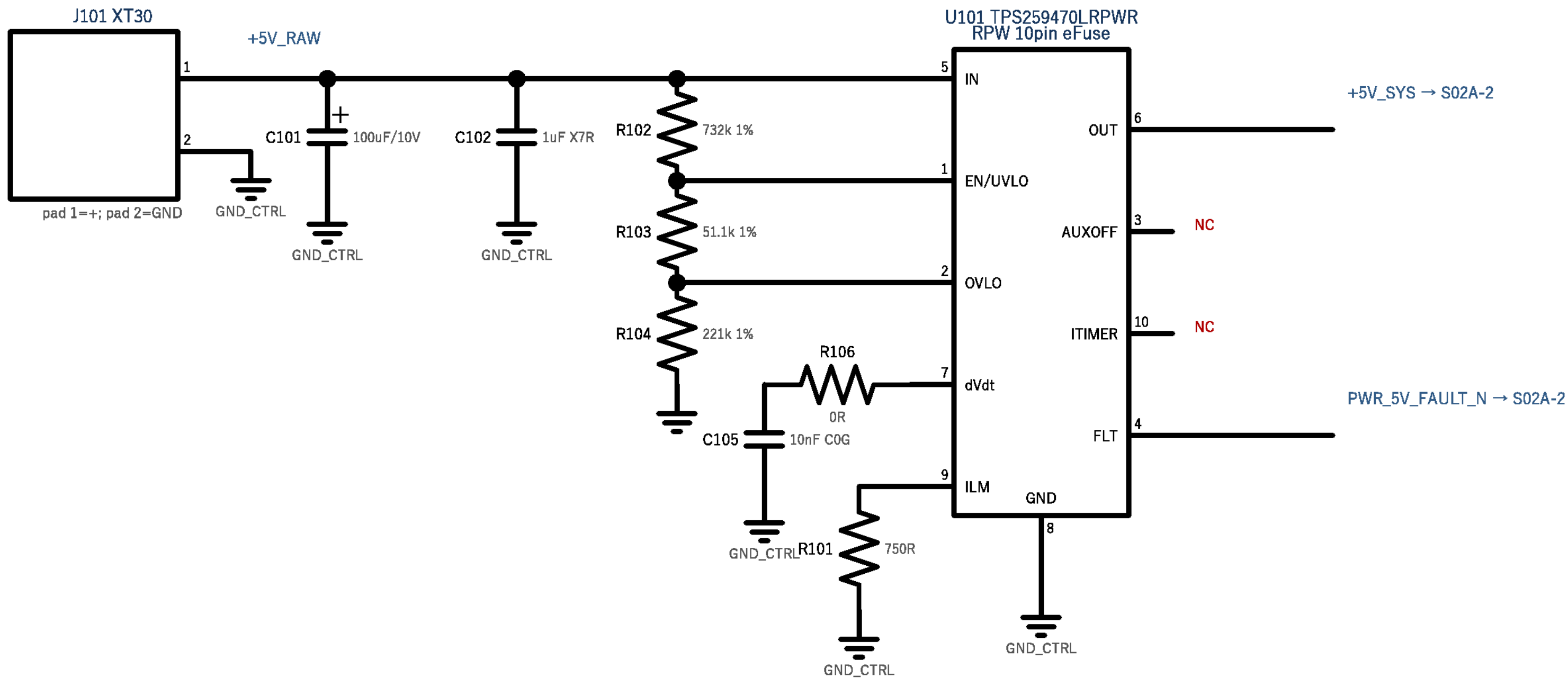
Priority: ADR + KiCad entry reference (2026-09-26 USB eFuse / GPIO VCC additions, pages 14-15). Existing 2026-09-17 PDF and legacy J402 driver text are superseded.

Dedicated CAN ports are horizontal JST GH3 (1=A, 2=B, 3=GND). Generic J323 remains 4pin.

No dedicated E228 main-contact feedback. Coil clamp/fuse and 24V return topology remain open on S11.

S02A-1 5V input and TPS259470 eFuse

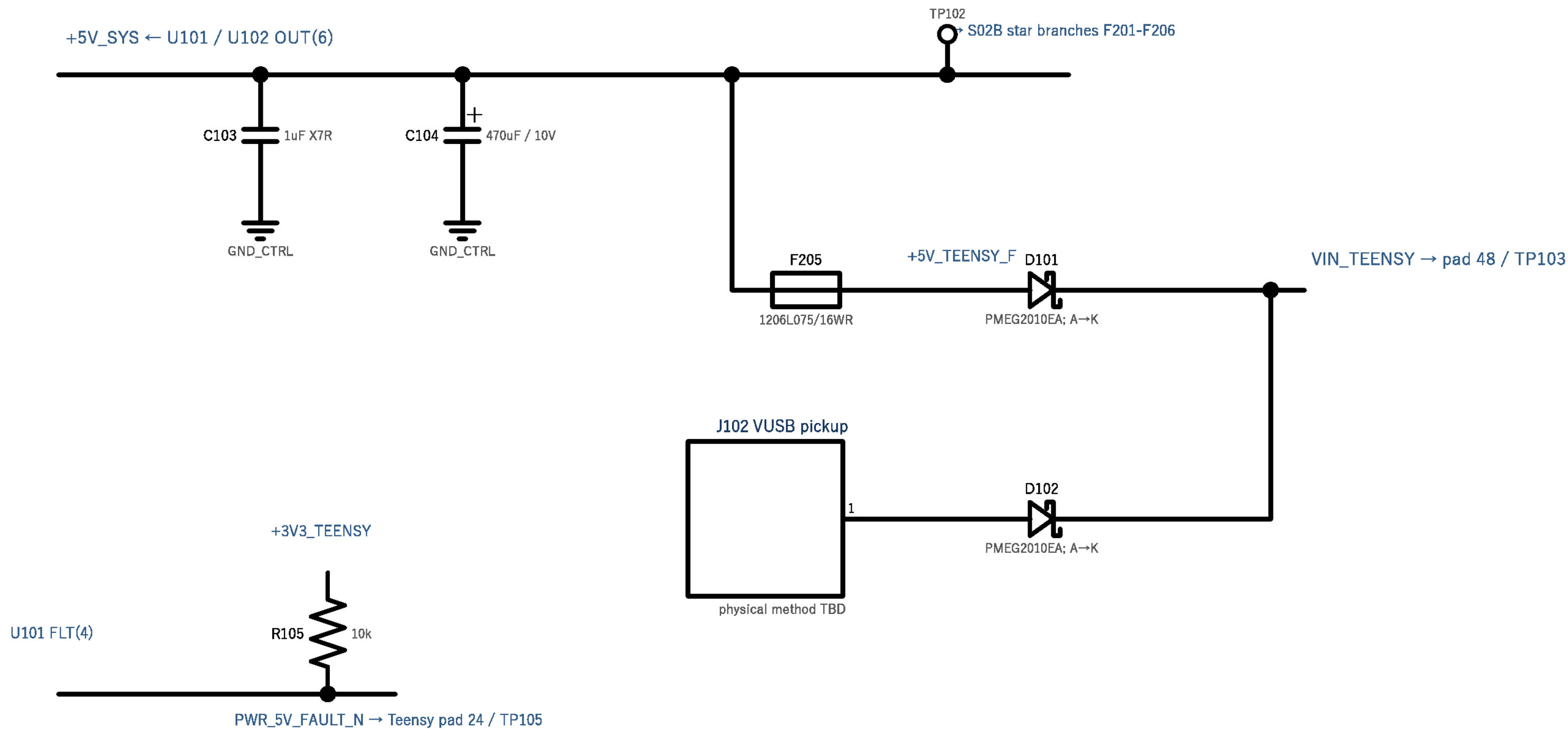
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Source: external SD-25B-5 → J101 XT30 → U101 TPS259470 → +5V\_SYS. There is no 24V-to-5V buck on this PCB.  
R102/R103/R104 form the UVLO/OVLO divider. U101 3=AUXOFF, 10=ITIMER are NC; 4=FLT has a 3.3V pull-up.

S02A-2 5V output and USB diode-OR to Teensy VIN

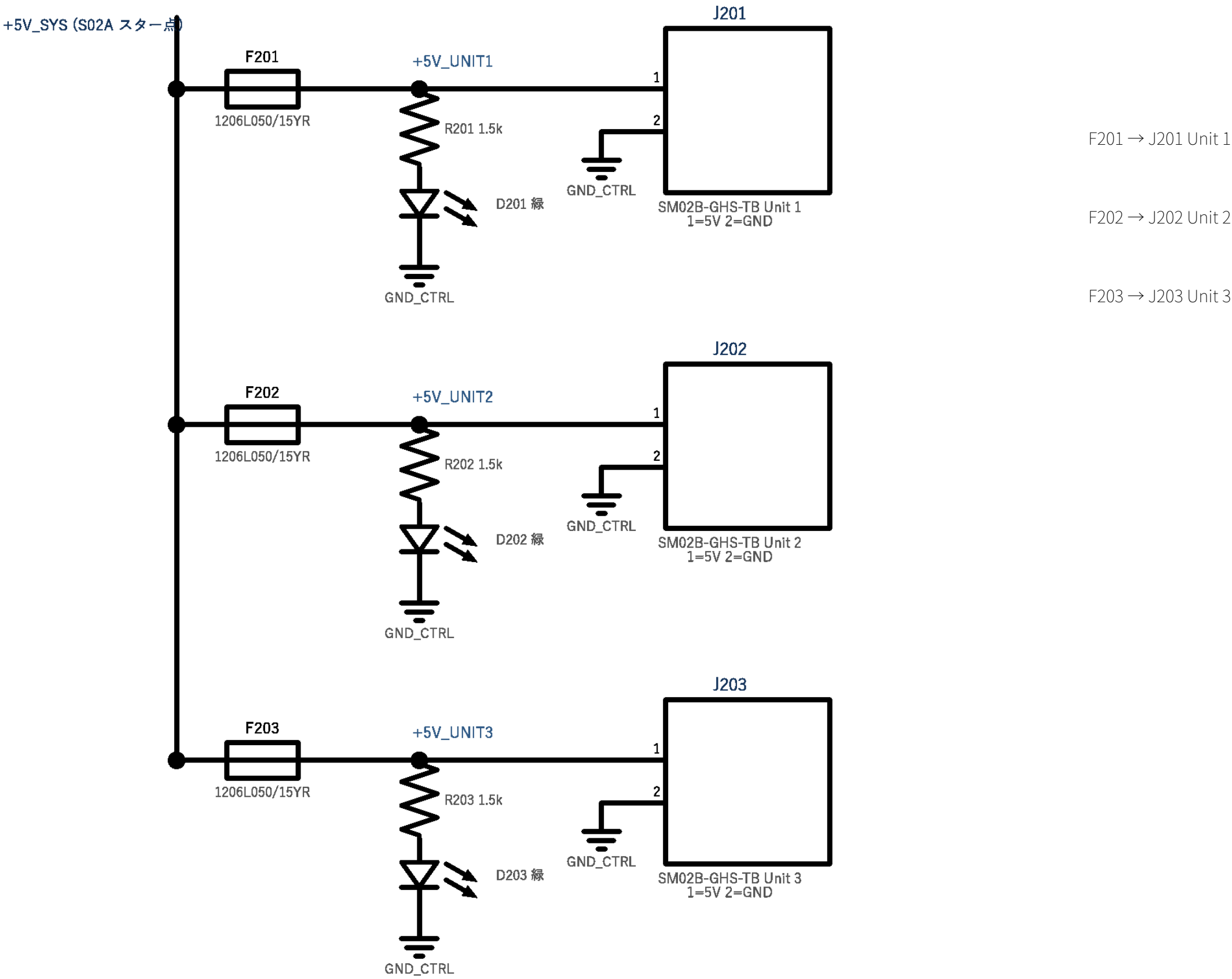
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D101 anode=F205 output; D102 anode=Teensy VUSB. Their cathodes join at VIN\_TEENSY / pad 48.  
Cut VUSB-VIN. J102 also feeds U102 IN(5) on S02A-USB (page 14); USB now supplies +5V\_SYS and one node.

# S02B-1 5V star branches to the three drive units

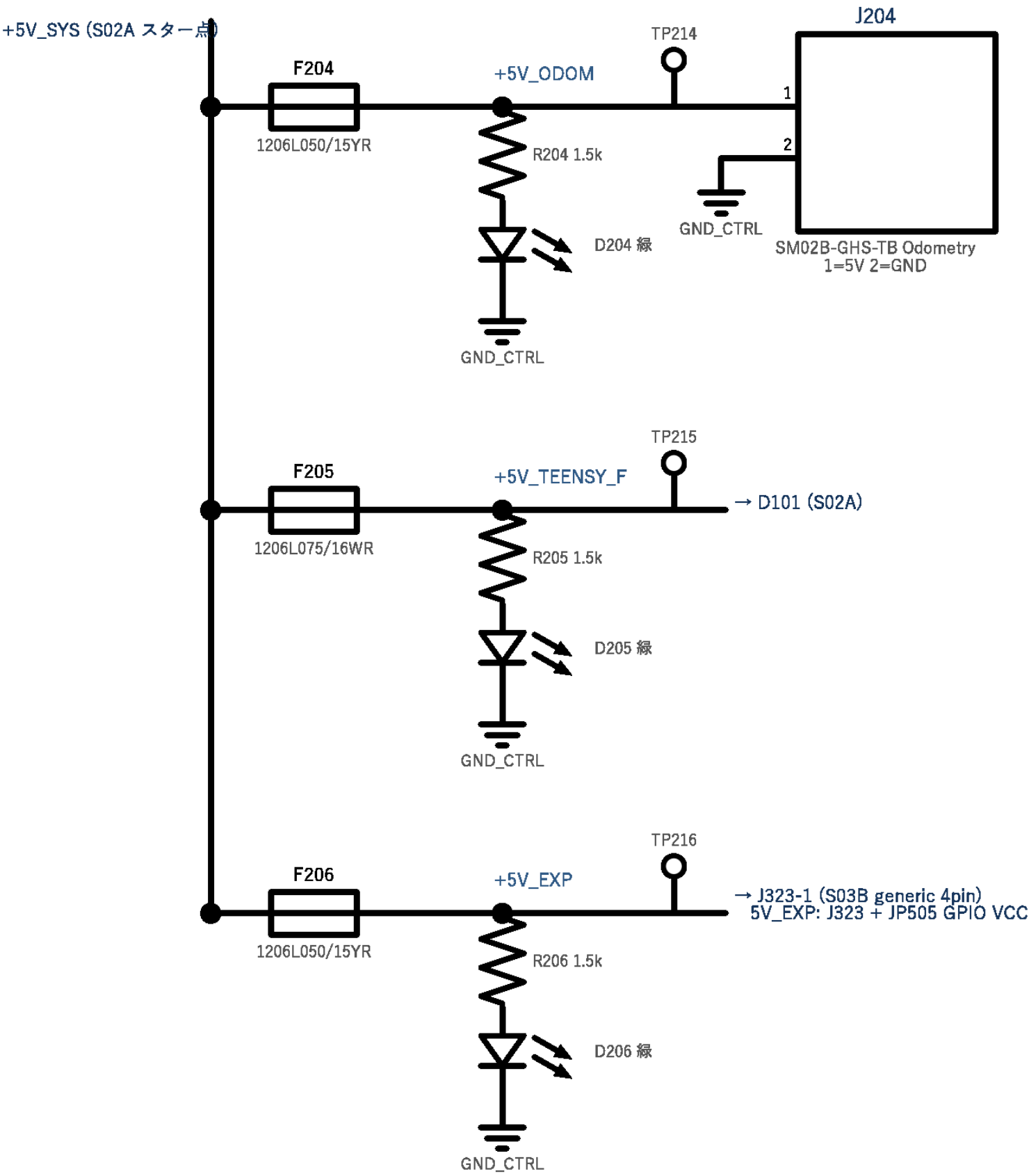
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J201-J203: pin 1=each protected 5V branch, pin 2=GND\_CTRL. CAN3 GH3 carries its own GND reference on pin 3.  
Unit TP211-TP213 omitted; measure at J201-J203 when needed. LEDs indicate power presence; keep protected nets separate.

# S02B-2 5V branches to ODOM, Teensy and generic expansion

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F204 → J204 ODOM

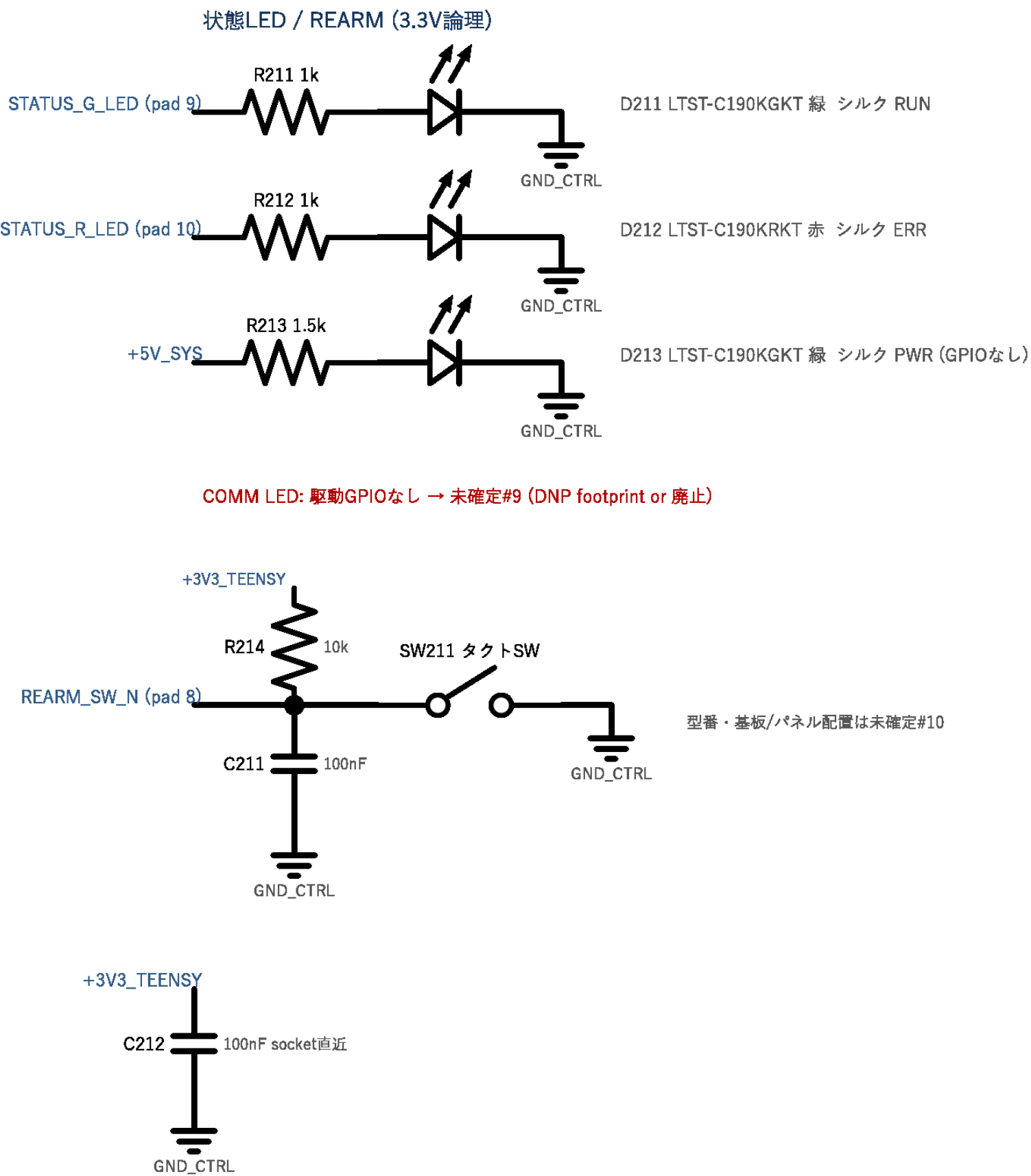
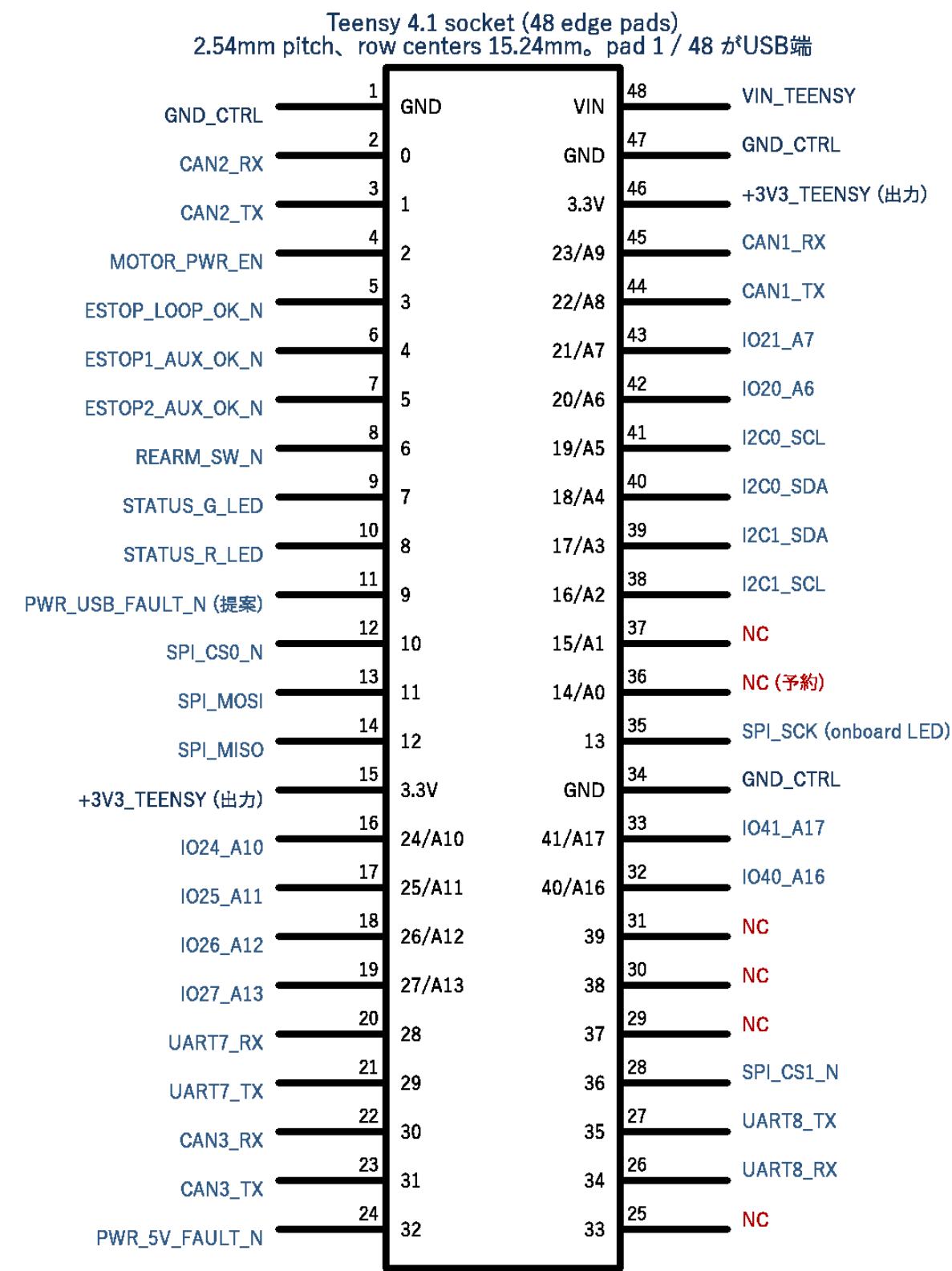
F205 → D101 Teensy VIN

F206 → J323 generic CAN2

J204 pin 2 carries ODOM supply GND. CAN1 GH3 also carries its GND reference on pin 3; F206 feeds generic J323 pin 1.  
F205/F206 hold ratings need measured current/inrush review. Do not power self-powered LED/switching nodes through J323 by assumption.

S02C Teensy 4.1 socket - all 48 edge pads and local indicators

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MOTOR\_PWR\_EN is pad 4 / Teensy pin 2 → U402 A(2). Pad 11 / pin 9 = PWR\_USB\_FAULT\_N (proposal); pad 36 / pin 14 remains NC.  
Teensy 3.3V pads 15/46 are outputs. Never back-feed them. Pin 13 has the on-board LED load and is SPI SCK.

# S02M Teensy socket placement - USB-end orientation

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PJRC mechanical drawing: 24 positions per row at 2.54 mm pitch; row center spacing is 15.24 mm.

17.78 mm is the Teensy board WIDTH, not the socket row spacing. Legacy central docs used it incorrectly.

Near/bottom row from USB end: Teensy pads 1 → 24 in order.

Far/top row from USB end: Teensy pads 48 → 25 in order.

USB end: pad 1 is left, pad 48 is right when viewed from component/top side.

Teensy socket footprint

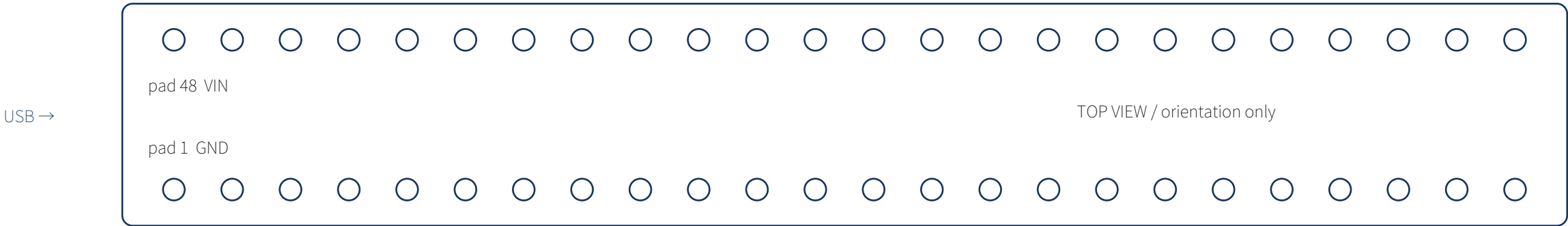
Prefer DifferentialSwerve:Teensy41\_Socket\_2x24  
48-pad combined footprint; matching 48-pin symbol required  
PJRC suggests Sullins PPPC241LFBN-RC / PPTC241LFBN-RC  
Verify socket lead fit; USB / Program / microSD access

Existing PCB placement warning

Current central-board.kicad\_pcb J3/J4 are about 33.3 mm apart and offset along the rows. They cannot mount a Teensy 4.1.  
Do not copy that placement into the new central layout.  
Keep components and exposed pads clear below the module.

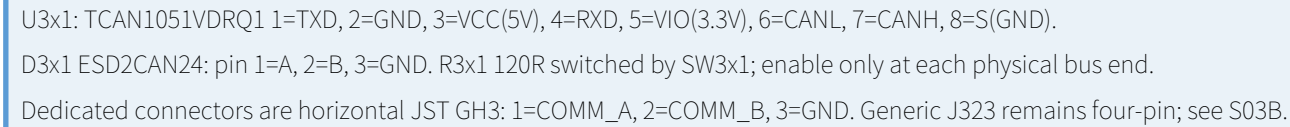
Before fab: compare KiCad footprint dimensions with PJRC official drawing, then print at 1:1 scale.

Check pad 4 → MOTOR\_PWR\_EN and both 3.3V/GND/VIN pads against the socket pin map on S02C.



Mechanical source: [https://www.pjrc.com/teensy/dimensions\\_teensy41.png](https://www.pjrc.com/teensy/dimensions_teensy41.png) ; PJRC socket: [https://www.pjrc.com/store/socket\\_24x1.html](https://www.pjrc.com/store/socket_24x1.html)

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# S03B CAN connectors and GND reference paths

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| Bus / nodes          | Central port  | Pin 1         | Pin 2         | Pin 3 / reference |
|----------------------|---------------|---------------|---------------|-------------------|
| CAN1 ODOM V2         | J311/J312 GH3 | SENSOR_COMM_A | SENSOR_COMM_B | GND_CTRL (pin 3)  |
| CAN2 LED / switching | J321/J322 GH3 | EXP_COMM_A    | EXP_COMM_B    | GND_CTRL (pin 3)  |
| CAN3 Drive V2 x3     | J331/J332 GH3 | DRIVE_COMM_A  | DRIVE_COMM_B  | GND_CTRL (pin 3)  |
| CAN2 generic         | J323 4pin     | +5V_EXP       | GND_CTRL      | 3=A / 4=B         |

CAN1: Teensy TX pin 22 / pad 44 → U311 TXD; RX pin 23 / pad 45 ← U311 RXD.

CAN2: Teensy TX pin 1 / pad 3 → U321 TXD; RX pin 0 / pad 2 ← U321 RXD.

CAN3: Teensy TX pin 31 / pad 23 → U331 TXD; RX pin 30 / pad 22 ← U331 RXD.

All ports use COMM\_A=CANH and COMM\_B=CANL. J323 stays a distinct four-pin powered generic port.

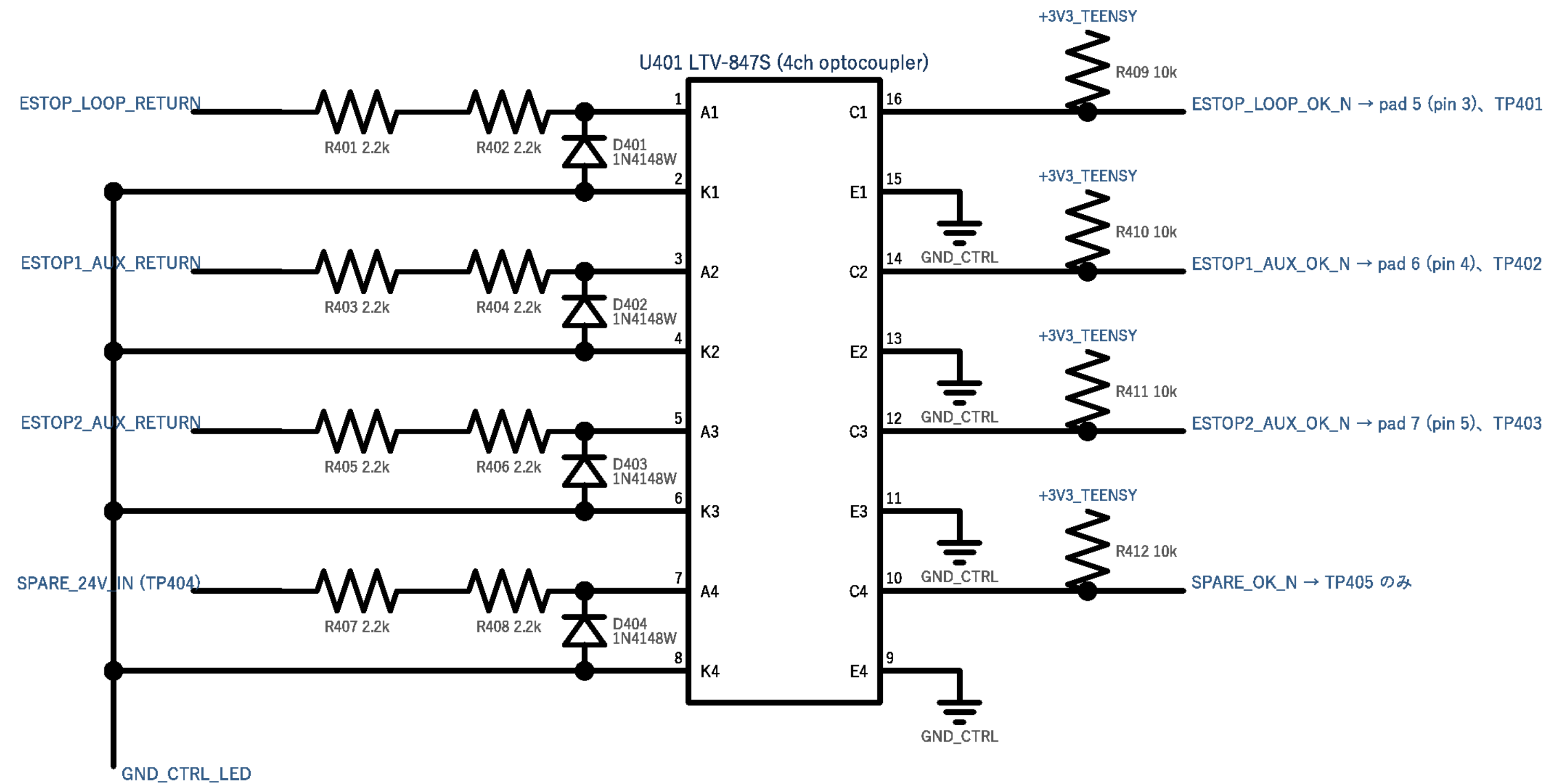
All dedicated CAN connectors: horizontal JST SM03B-GHS-TB; pin 1=A, 2=B, 3=GND. 5V/GND power wiring stays separate.

CAN reference GND must not carry LED/motor load current. Keep the three CAN systems separate.

Verify horizontal GH3 footprint pin-1 orientation, mating housing/contact and cable view. GH2 supply remains a separate connector.  
J323 is a distinct powered four-pin generic CAN port; do not copy its pin order onto dedicated GH3 ports.

S04A E-stop NC chain and LTV-847S diagnostic monitoring

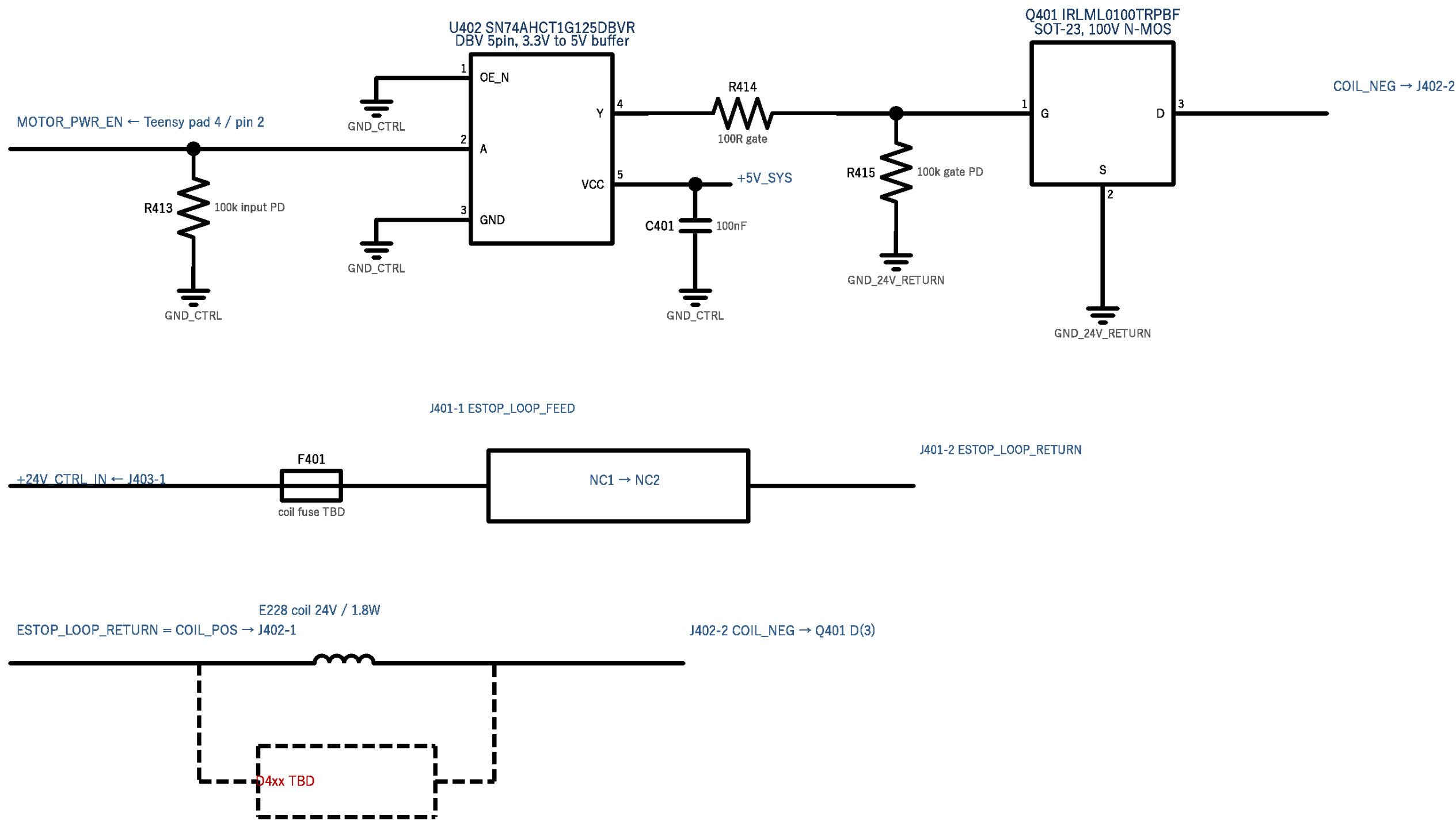
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J401 ESTOP\_CTRL: 1=LOOP\_FEED, 2=LOOP\_RETURN, 3=+24V\_CTRL\_LED, 4=GND\_CTRL\_LED, 5/6=AUX1/AUX2\_RETURN.  
J403-1 → fuse TBD → J401-1 → NC1 → NC2 → J401-2 → COIL\_POS/J402-1. This is a physical coil path independent of firmware.  
U401 ch1 LED 1/2 ↔ transistor 16/15; ch2 3/4 ↔ 14/13; ch3 5/6 ↔ 12/11; ch4 7/8 ↔ 10/9.  
Each 1N4148W is reverse parallel to its opto LED (anode at K). AUX1/2 monitor E-stop buttons, not E228 main contacts.

# S04B Central PCB integrated E228 coil driver

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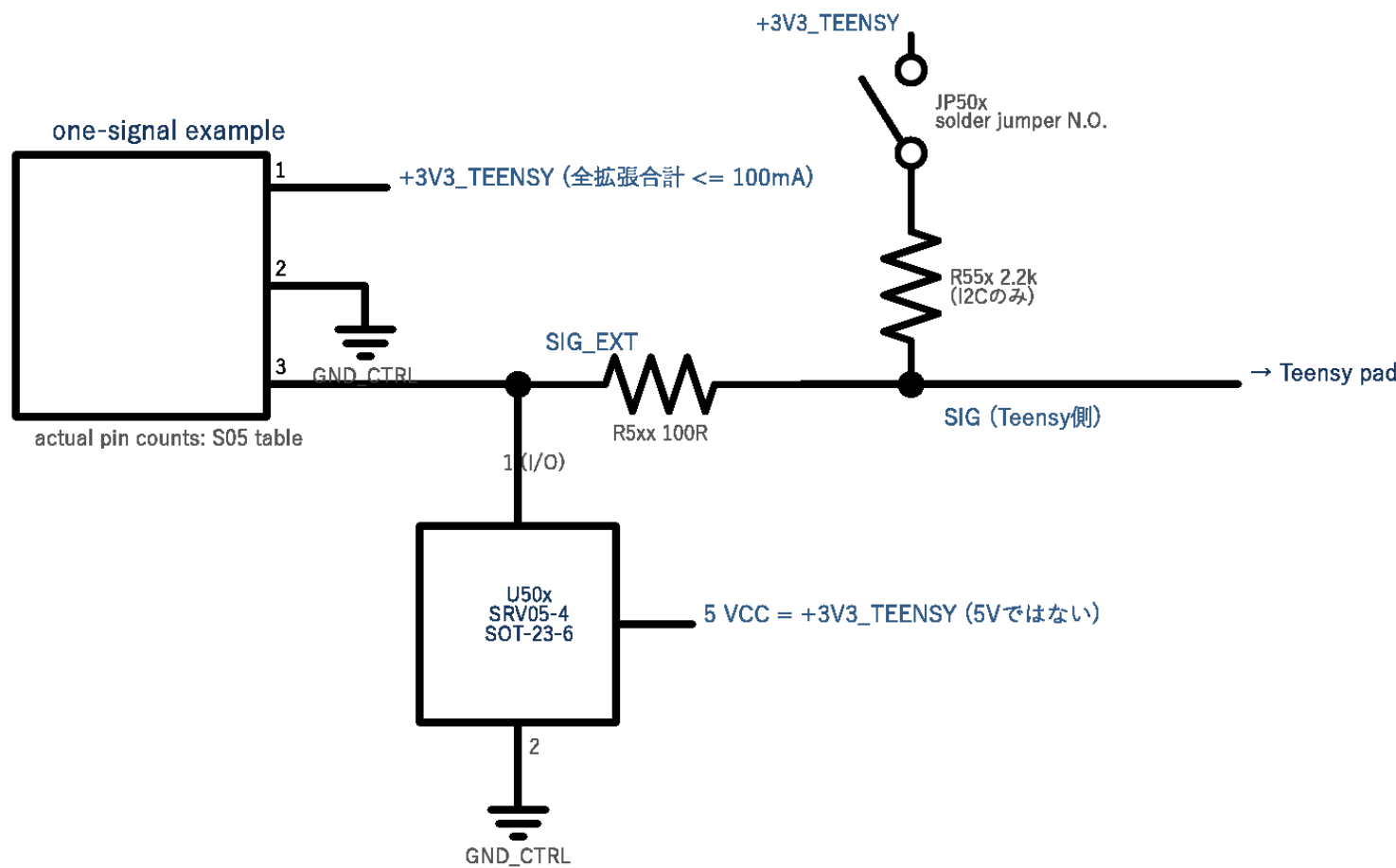


Dashed D4xx is NOT WIRE. STOP: clamp topology/value/polarity and release time TBD

Candidate U402 DBV pins: 1=OE\_N tied low; 2=A from Teensy; 3=GND; 4=Y; 5=+5V\_SYS. R413/R415=100k pull-downs; R414=100R. Q401 IRLML0100 pin 1=G, 2=S, 3=D. J403-2=GND\_24V\_RETURN; J402-2=COIL\_NEG. NC loop cuts COIL\_POS even if Q401 is on. F401 value/location, flyback clamp type/orientation, release time, J402/J403 MPN and GND bond are STOP/TBD before KiCad release. No dedicated main-contact feedback. C620 CAN response loss is only auxiliary diagnostic, never proof that E228 contacts opened.

S05 Protected external I/O common circuit

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| Ref       | Function / connector order                                     |
|-----------|----------------------------------------------------------------|
| J501      | Matek I2C 4pin: GND / SDA / SCL / +5V_SYS                      |
| J502/J503 | I2C0 / I2C1: 3V3 / GND / SDA / SCL                             |
| J504/J505 | UART7 / UART8: GND / TX / RX                                   |
| J506      | SPI 7pin: 3V3 / GND / SCK / MOSI / MISO / CS0_N / CS1_N        |
| J507      | GPIO/ADC 10pin: VCC_GPIO_EXT / GND / IO20/21/24/25/26/27/40/41 |

SRV05-4 x6: pins 1/3/4/6=protected lines, pin 2=GND, pin 5=+3V3\_TEENSY.

Every external signal: connector → TVS → 100R series → Teensy. I2C pull-ups are jumper-selectable.

J507 VCC is selected by JP505 (page 15). Signals and TVS pin5 stay 3.3V; I2C/SPI VCC stays 3.3V.

Connector form factor and Matek +5V branch protection remain as earlier open items. Total external 3.3V load <=100mA.

# S11 Open design decisions and manual KiCad checks

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CHECK 1 Dedicated CAN horizontal JST GH3 pin 1=A, 2=B, 3=GND; verify footprint, mate and cable orientation.

STOP 2 Coil fuse F401 rating/location and LED branch protection; confirm 24V supply maximum.

STOP 3 Coil flyback/clamp topology and part; measure E-stop release time and Q401 VDS peak.

STOP 4 J402 coil/J403 24V pin order, MPN and keying; verify E228 coil polarity if clamp is polarized.

STOP 5 5V GND, 24V coil return and CAN reference GND bond; ensure coil return does not cross quiet GND.

STOP 6 Teensy socket drill/pad/mechanical clearance and VUSB pad pickup method.

STOP 7 J501 Matek pull-up voltage; F205/F206 current and inrush; U401 package/land pattern.

STOP 8 Confirm U402 OE\_N tie and power-up/down behavior on hardware; Reset/Hi-Z must keep Q401 off.

Transcription and review sequence

1 Reconcile every connector pin and every IC pin with this PDF, manufacturer pin diagrams and the final harness drawing.

2 Check socket orientation and 15.24 mm row spacing by 1:1 print; ignore current PCB J3/J4 placement.

3 Verify physical NC continuity from J403 via both E-stops to coil +; open either NC and measure coil de-energize.

4 Run KiCad ERC, then compare schematic netlist pin-by-pin to PCB pads; inspect all power pads and NC markers.

5 Run DRC and perform powered bench test: USB only, 5V only, combined, E-stop, reset, CAN GND under load.

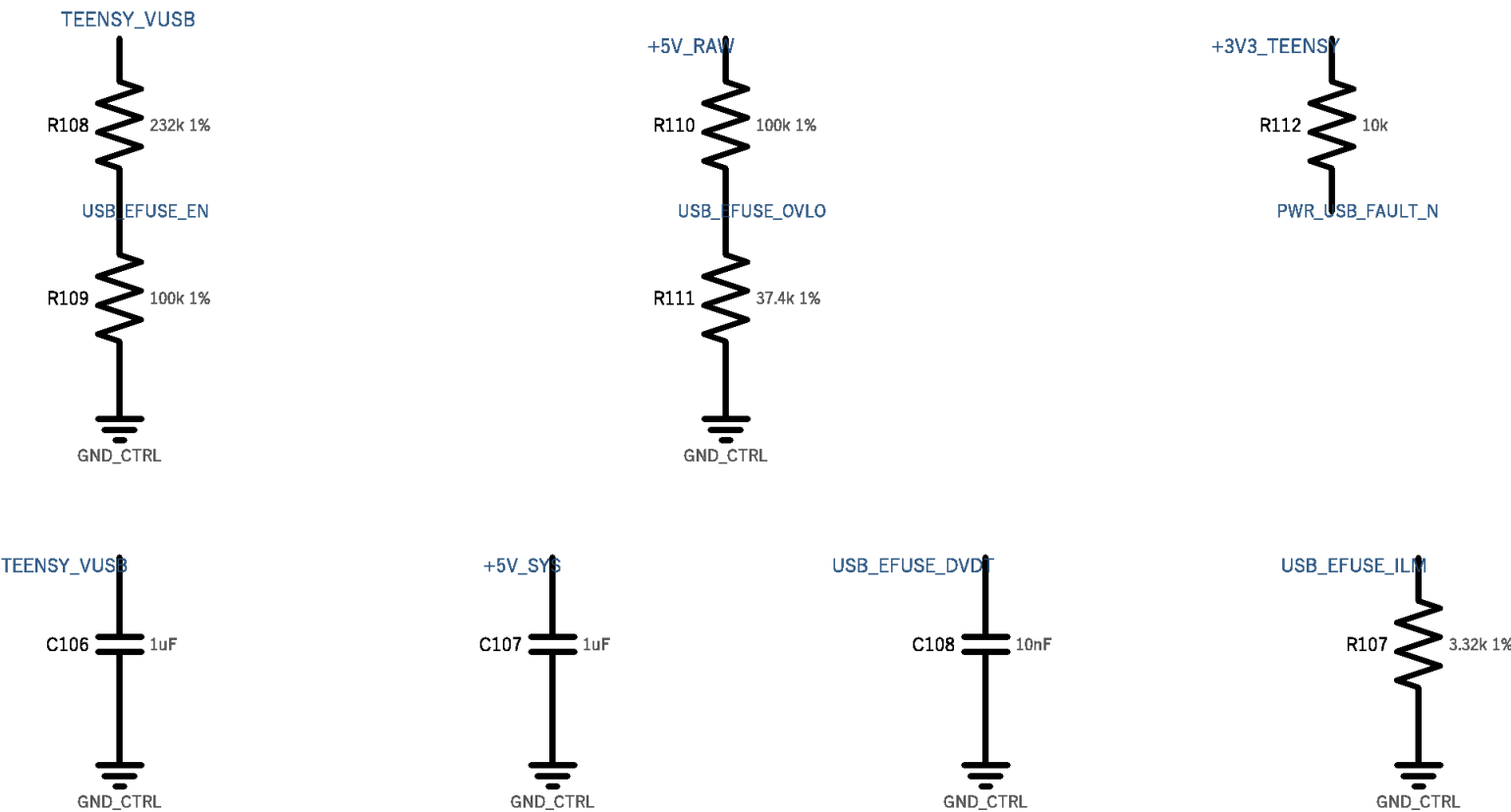
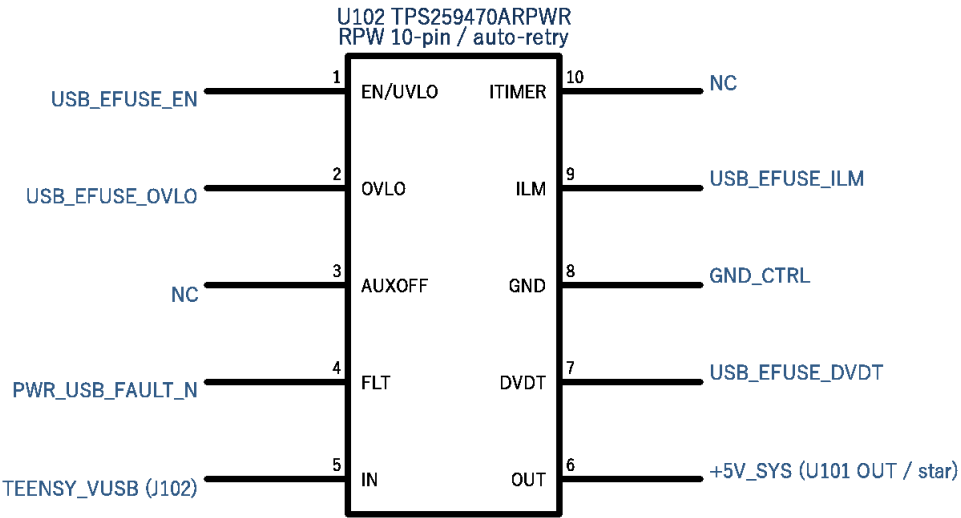
No main-contact monitor: when OFF command is issued and C620 replies cease, display OFF指令中・C620応答なし. CAN silence alone must never mean 遮断確認済み.

If central↔drive CAN itself is unavailable, show unknown/stale state, distinct from confirmed C620 nonresponse reported by drive MCU.

Legacy 2026-09-17 PDF has obsolete J402 external-driver text. Keep hard NC coil cut, reset/Hi-Z OFF and explicit re-arm.

# S02A-USB USB eFuse - CAN and one 5V node without external power

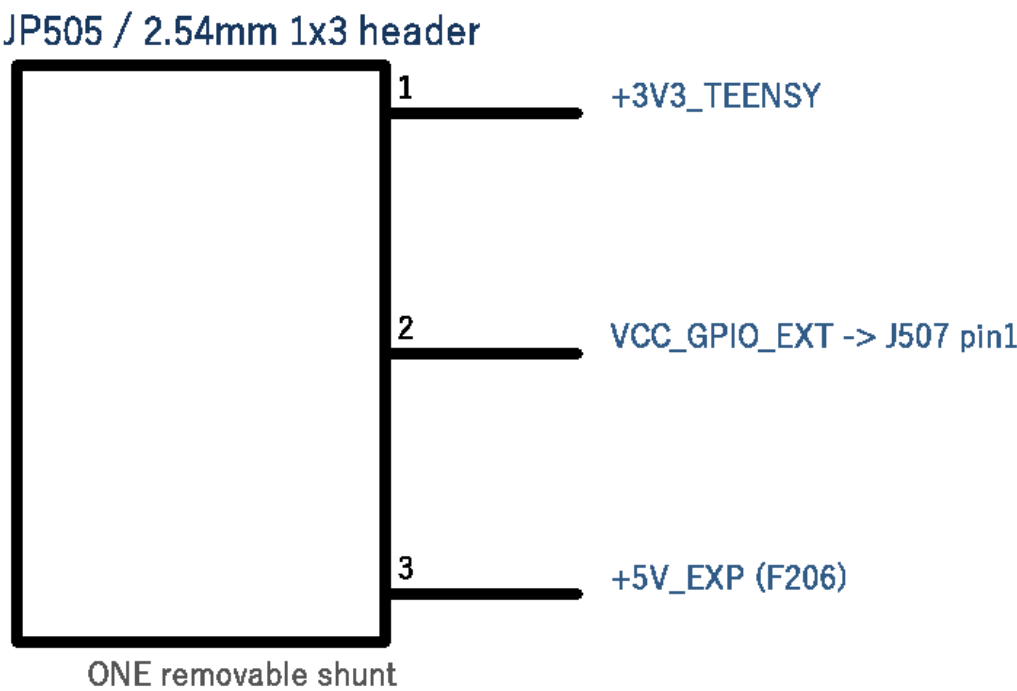
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U102 IN5=J102 VUSB; OUT6=+5V\_SYS. U101 / U102 true RCB blocks reverse current; D102 direct VIN path remains.  
TP106=PWR\_USB\_FAULT\_N -> Teensy pad11 / GPIO9 (proposed). R112 pull-up goes to +3V3\_TEENSY.  
R107 3.32k: ILM 0.85 / 1.007 / 1.15A. R110/R111: external 5V >=4.41V switches U102 OFF (nominal).  
U102 RPW footprint, VUSB pickup/current capacity and loaded +5V\_SYS >=4.5V remain to be checked.  
TI TPS25947 SLVSFC9C Rev.C; entry reference 3.1. USB3 / powered hub recommended; measure actual load.

S05-VCC GPIO supply selector - one header for all GPIO VCC pins

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1-2 = 3.3V (default)  
2-3 = 5V  
Only supply pin changes. IO stays 3.3V.

JP505 pin1=+3V3\_TEENSY; pin2=VCC\_GPIO\_EXT; pin3=+5V\_EXP. J507 pin1 connects to JP505 pin2.

Use one shunt: 1-2 = 3.3V default; 2-3 = 5V. Header/shunt purchase MPN remains unspecified.

Teensy IO remains 3.3V ONLY. U505/U506 SRV05-4 pin5 stays +3V3\_TEENSY; ADC function remains.

I2C J502/J503 and SPI J506 supply pins remain fixed at 3.3V. Total external 3.3V load <=100mA.

5V GPIO supply shares F206 with J323; review combined current. Silk: VCC 3V3/5V and IO 3.3V ONLY.